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SECTION: E

Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM #27

Warehouse Inventory & Pallet Location Tracker

Deliverable 1: Requirements Table

FUNCTIONAL REQUIREMENTS (FR)

Req ID	Type	Priority	Description	Acceptance Criteria	Rationale
FR-001	Functional	High	Pallet Placement Validation Against Weight Capacity: The system shall validate proposed pallet placement coordinates (X, Y, Z) against the rack's maximum load capacity before confirming storage.	Pass: Pallet assigned to valid rack when cumulative weight $\leq$ rated capacity; system confirms placement and logs transaction. Fail: Placement rejected if cumulative weight $>$ rated capacity; system displays error message with available capacity details. Pass: System provides alternative rack suggestions with sufficient capacity	Exceeding weight limits poses structural safety risks. This is a critical constraint enforced before pallet storage.
FR-002	Functional	High	Barcode/RFID Scanning and Item Identification: The system shall support scanning pallet barcodes and RFID tags to uniquely identify warehouse items. When a barcode/RFID is scanned, retrieve associated item metadata (SKU, product name, quantity, weight, total	Pass: Valid barcode/RFID scan retrieves correct item metadata within 2 seconds. Pass: Duplicate scan detection prevents duplicate entries; system alerts operator. Fail: Invalid or unregistered barcode triggers error message; system logs failed attempt with timestamp. Pass: Barcode/RFID data is case-insensitive and	Automated identification via barcode/RFID is essential for accurate inventory tracking, preventing manual errors, and maintaining real-time visibility.

			pallet weight) for operator verification.	tolerates minor read inconsistencies	
FR-003	Functional	High	3D Pallet Location Tracking and Retrieval: The system shall store and manage pallet location data in 3D coordinates (X, Y, Z) representing shelf rack positions. Enable operators to query pallet location by SKU or barcode and retrieve precise 3D coordinates for efficient warehouse navigation and retrieval.	Pass: Pallet location stored with X, Y, Z coordinates accurate to within 1 cm. Pass: Location query by SKU returns correct 3D coordinates within 100 ms for warehouses up to 100,000 bins. Pass: Multiple pallets of same SKU individually tracked with unique location identifiers. Pass: System prevents coordinate conflicts (two pallets cannot occupy same location)	Precise 3D location tracking enables quick pallet retrieval, reduces picking time, and minimizes search errors. Performance ensures scalability for large warehouses.
FR-004	Functional	Medium	Stock Movement Logging and Audit Trail: The system shall automatically log all stock movements (pallet inbound, outbound, relocation) with timestamp, operator ID, source location, destination location, barcode/RFID scanned, and action type. Audit trail shall be immutable and available for compliance and inventory reconciliation.	Pass: Every stock movement creates immutable log entry within 500 ms of action completion. Pass: Audit trail includes timestamp, operator ID, action type, pallet ID, source location, destination location, status. Pass: Historical audit data retained minimum 2 years and accessible via filtered queries. Pass: System prevents deletion or modification of historical log entries	Comprehensive audit trails required for compliance, inventory reconciliation, loss prevention, and dispute resolution. Immutability ensures data integrity for regulatory audits.
FR-005	Functional	Medium	Real-Time Warehouse Inventory Status Dashboard: The system shall provide	Pass: Dashboard displays current occupancy and capacity for all racks with refresh	Real-time visibility enables warehouse supervisors to make informed decisions about inventory

			real-time dashboard displaying current warehouse inventory status including total pallets stored, available storage capacity, occupancy percentage per rack/shelf, and alerts for racks approaching maximum weight capacity. Dashboard shall update automatically when pallets are added or removed.	rate ≤ 5 seconds. Pass: Racks with $>80\%$ weight capacity highlighted with visual warning. Pass: Dashboard shows total active pallets, total SKUs, and available storage slots. Pass: Operators can filter dashboard view by rack zone, aisle, or product category. Pass: Dashboard accessible via web browser and mobile app with same data freshness	distribution, capacity planning, and operational efficiency. Visual alerts prevent accidental overweight situations.
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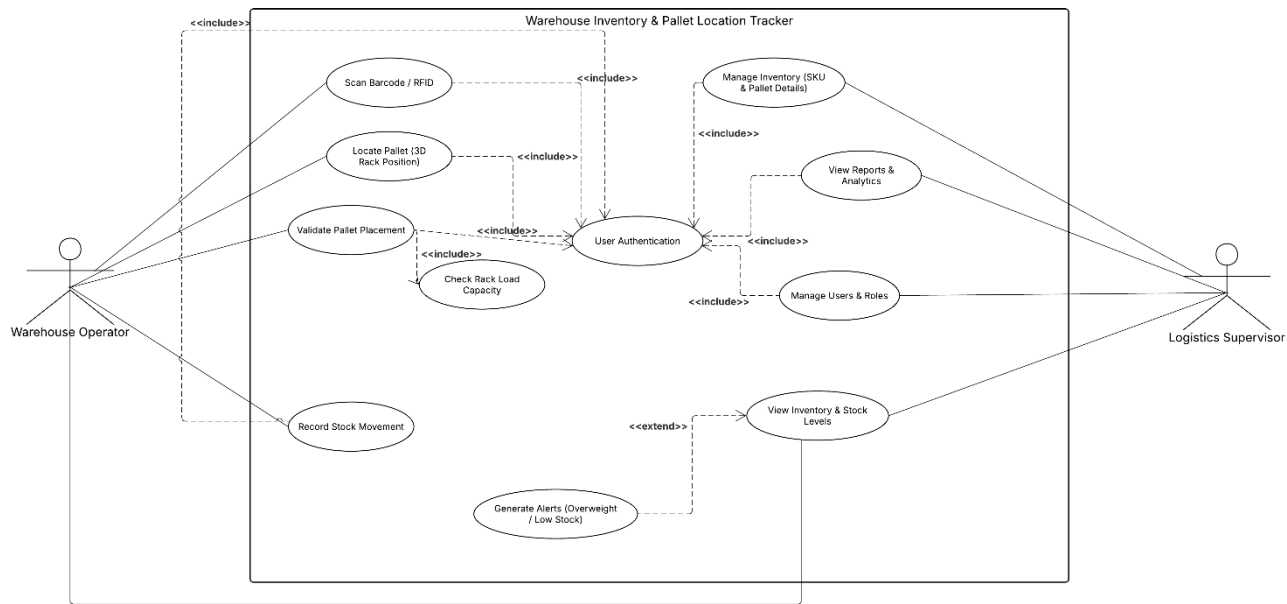
NON-FUNCTIONAL REQUIREMENTS (NFR)

Req ID	Type	Priority	Description	Acceptance Criteria	Rationale
NFR-001	Non-Functional (Performance)	High	Performance and System Response Time: The warehouse item lookup service must locate any SKU pallet coordinate across up to 100,000 bins and return results in under 100 ms under normal operating conditions. Barcode/RFID scanning and	Pass: Benchmarking tests confirm SKU lookup latency ≤ 100 ms for 100,000-bin warehouse under single and concurrent load. Pass: Barcode scan processing + weight validation completes within 2 seconds in 99th percentile. Pass: System maintains response time targets when 50	Warehouse operations require fast response times to minimize operator wait time, reduce bottlenecks at receiving/dispatch, and ensure smooth workflow. Concurrent user support essential for multi-operator facilities.

			weight validation queries must complete within 2 seconds. System shall support concurrent operations for up to 50 warehouse operators simultaneously without performance degradation.	concurrent operators are active. Pass: Database query execution plans optimized with indexed searches on SKU, barcode, and coordinates. Fail: If average response time exceeds 200 ms or 99th percentile exceeds 3 seconds	
NFR-002	Non-Functional (Security)	High	Data Security and Access Control: The system shall implement role-based access control (RBAC) restricting warehouse operators to view/modify only inventory relevant to their assigned zones/shifts. Sensitive data (pallet locations, inventory counts, audit logs) shall be encrypted at rest using AES-256 and in transit using	Pass: RBAC policies restrict operators to assigned zones; unauthorized access attempts logged and denied. Pass: All data at rest encrypted with AES-256; encryption keys managed by secure key management service. Pass: All data in transit uses TLS 1.2+ with valid certificates; network packet inspection confirms encryption. Pass: All user logins audited with	Inventory data represents significant business asset value. Security breaches could lead to theft, unauthorized inventory manipulation, or operational disruption. Encryption and RBAC are industry-standard practices for warehouse management systems.

			TLS 1.2 or higher. All user authentication shall use OAuth 2.0 or industry-equivalent with multi-factor authentication (MFA) for supervisor-level accounts.	timestamp, user ID, and success/failure status. Pass: Supervisor and admin accounts require MFA (TOTP or hardware token); enforcement verified. Fail: If unencrypted sensitive data transmitted or stored, or if unauthorized access not prevented/logged	
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Deliverable 2: UML Use-Case Diagram



Deliverable 3: Use-Case Flow Specification

Use Case: UC-03 – Validate Pallet Placement

Primary Actor: Warehouse Operator

System: Warehouse Inventory & Pallet Location Tracker

1. Preconditions

1. The Warehouse Operator has access to the Warehouse Inventory & Pallet Location Tracker.
2. The pallet has been identified through Barcode/RFID scanning.
3. The proposed 3D rack/pallet placement coordinates are available.
4. The selected rack has a defined maximum load capacity.

2. Postconditions

Success:

- The proposed pallet placement is validated successfully.
- The pallet is assigned to a valid rack position.
- The pallet storage information is recorded in the system.

Failure:

- The pallet is not placed on the selected rack when the total weight exceeds the rack's rated structural limit.
- The system informs the Warehouse Operator that the placement is invalid.

3. Main Success Scenario

1. The Warehouse Operator scans the pallet using Barcode/RFID.
2. The system identifies the pallet and retrieves its relevant inventory information.
3. The Warehouse Operator enters or selects the proposed 3D rack position.
4. The system checks the selected rack and retrieves its maximum load capacity.
5. The system calculates the total weight that would be present on the rack after placement.
6. The system compares the calculated total weight with the rack's maximum load capacity.
7. The system confirms that the rack can safely accommodate the pallet.
8. The system validates the pallet placement.
9. The system assigns the pallet to the selected rack position and records the storage information.

4. Alternate Flow – Rack Capacity Exceeded

A1. At Step 6, the system determines that the total weight after pallet placement exceeds the rack's rated structural limit.

A2. The system rejects the proposed pallet placement.

A3. The system displays an alert indicating that the selected rack has insufficient load capacity.

A4. The Warehouse Operator selects another valid rack position.

A5. The system repeats the validation process for the newly selected rack.

Acceptance Criteria

Pass: The pallet is assigned to a valid rack.

Fail: Placement is not permitted on a rack where the total weight exceeds the rated structural limit.